

**The Flat-Line Theorem:
the unique solution manifold
through the critical point**

$F = r = V = 0$ exactly on the line · b_2 is static, but the flow leaves S : $|\text{Dr}V| = 39.2 \cdot A$

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Abstract

This monograph records the main result of the nonlinear DAE march: the unique solution manifold of the de-adiabatized truncated tower through the critical point is the flat line of equilibria. The nonlinear prolongation is built by the chain rule without linearization, the consistency of the closure gives nonlinear hidden conditions, and three machine theorems close the question at two points ($\kappa = 2$ and $\kappa = 2 - \pi^2/98$): the flat line carries $F = r = V = 0$ exactly and marches eight echoes with zero drift; the direction e_2 is statically empty; the second direction b_2 holds the radius statically ($|F| \leq 6 \times 10^{-14}$), but the flow immediately leaves the admissible set with the rate $|DrV| = (39.2 \pm 0.2) \cdot A$, linear in the amplitude, with a breakdown in the first RK4 stage.

The corollaries: the Jordan-2 linear dynamics is the linearization of no nonlinear flow; the “baryon asymmetry” annihilation continues on the nonlinear level; the near-critical physics remains with the PDE machine. The march and exit tables, the machine statistics (11 755 closures, 0 rank drops), the honest caveats and the reproduction protocol are included.

Keywords: flat line, nonlinear DAE, hidden conditions, branch exit, Jordan-2, machine proof, Choptuik problem.

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1. Introduction

The sixth monograph closes the Russian (exact) block of the series with the flat-line theorem — the main claim of the nonlinear DAE march of campaign v15 (machine `march_dae_nonlinear.py` on the core `dae_nonlinear_core.py`, LEDGER §26). The linear analysis of v14 left one question open: the world of the linearization annihilated exactly ($\delta_{\text{mono}} = 0$, Jordan-2), but linear artifacts often do not continue to the nonlinear level. Theorem 6 answers exhaustively: the unique solution manifold of the nonlinear de-adiabatized DAE through the critical point is the flat line of equilibria; a nonlinear march off the line is algebraically impossible.

The proof rests on three machine theorems T1–T3 executed at two points (the baseline $\kappa = 2$ and the one-brick $\kappa = 2 - \pi^2/98$): the flat line carries $F = r = V = 0$ exactly and marches eight echoes with zero drift; the direction e_2 is statically empty; the second null direction b_2 holds the radius statically, but the flow immediately leaves the admissible set with a rate linear in the amplitude. Jointly: consistent finite-amplitude nonstationary initial data near x^* are empty; the Jordan-2 linear dynamics is the linearization of no nonlinear flow.

2. Notation and Setting

The nonlinear DAE of the system: $F(x) = 0$ — twelve tower equations plus six chain-prolongation equations; the state x contains the amplitudes and their derivatives. The prolongation closure $U_k(x) \cdot u = -\text{Known}(x)(v, dd)$ is built by the chain rule $dF/d\tau = 0$ without linearization, with x -dependent Jacobians; the consistency of the closure $r(x) = 0$ gives the nonlinear hidden conditions (the left kernel of U_k , Procrustes alignment, invariant under the choice of basis). The admissible set is $S = \{F = 0\} \cap \{r = 0\}$.

The setting: classify the solutions of S in the neighbourhood of the critical point x^* . More precisely: (T1) prove that the flat line (the a -pure direction of $\ker B$) consists of exact equilibria with $F = r = V = 0$; (T2) determine whether other null directions of the linear level touch S ; (T3) if a branch of S exists, does it carry solutions — that is, does the flow continue from it. All three clauses are executed by the machine at both points; the cross-check of the Jacobians against the exact session-17 matrices gives 10^{-15} – 10^{-16} .

3. Statement of the Theorem

Theorem 6 (the flat line as the unique solution manifold). Let x^* be the critical point and $S = \{F = 0\} \cap \{r = 0\}$ the solution set of the nonlinear DAE. Then at both points ($\kappa = 2$, $\kappa = 2 - \pi^2/98$):

(T1) the flat line — the a-pure direction of $\ker B$ ($v = dd = 0$, the chain run R1h–R3h–R5h–M5h) — consists of exact equilibria: $F = r = V = 0$ with machine zero ($\max|F| \leq 6.3 \times 10^{-13}$ over all probes $s \in \{0.001 \dots 1\}$); the march of eight echoes gives the drift 0.0 (baseline) and 8.6×10^{-12} (brick) — the line is frozen exactly;

(T2) the statics of S : the direction e_2 (the linear M2 block) does NOT touch S — the cokernel ratio 0.0577 is constant in h , the joint Newton collapses $0.265 \rightarrow 0.081$; but the second null direction b_2 (the share $t_3 = 0.985/0.972$) exists: the b_2 -branch holds the radius statically ($\times 1.03\text{--}1.04$, $|F| \leq 5.9 \times 10^{-14}$, $|r| \leq 1.4 \times 10^{-15}$) up to $|x - x^*| \approx 1.35$;

(T3) the flow from the b_2 -branch is consistent at the point but immediately leaves S : the level-3 exit rate

$$|DrV| = (39.2 \pm 0.2) \cdot A, \quad \text{linear in the amplitude } A \quad (6.1)$$

— the closure breaks down in the very first RK4 stage ($|r| = 3.93 \times 10^{-5}$ at $A = 0.001$; 3.92×10^{-4} at $A = 0.01$): the b_2 -branch carries no solutions.

Outcome. The unique solution manifold through the critical point is the flat line of equilibria. The linear M2 dynamics (Jordan-2, $\delta_{\text{mono}} = 0$) is an artifact of linearization: it linearizes no nonlinear flow. The “baryon asymmetry” annihilation continues on the nonlinear level: the derived second-flow dynamics produces no finite-amplitude residual in the neighbourhood of x^* .

4. Sketch of the Machine Proof

4.1. The nonlinear core and the machine statistics

The core `dae_nonlinear_core.py` lambdifies the full nonlinear $F(x)$ with x -dependent Jacobians; the cross-check against the exact session-17 matrices gives relative deviations $10^{-16}\text{--}10^{-15}$ (J_a), $10^{-16}\text{--}10^{-14}$ (J_v), 0.0 (J_2). The prolongation is built by the chain rule

without linearization; the hidden conditions are the left kernel of U_k with Procrustes alignment (a norm-invariant basis). The machine statistics: 11 755 closure assemblies per run, zero rank drops, min rank = 9, maximum solver residual 3.9×10^{-4} — all breakdown landings are localized in the first stage of the march (clause T3).

4.2. The three steps

Step T1: along the line (parameter s) the machine checks F , r , V and the solver; all four quantities are machine zeros; the RK4 march of eight echoes with the projections $\{F = 0\}/\{r = 0\}$ does not change the distance to the line (13.4807 per echo, drift 0.0). Step T2: for $e2$ — the cokernel ratio is constant in the kick amplitude ($0.05766 \rightarrow 0.05793 \rightarrow 0.05793$), the joint Newton on $[F; r]$ contracts the radius without finding a branch; for $b2$ — the branch exists: the cokernel ratio 10^{-6} – 10^{-7} , the radius held (ratio 1.0255/1.0383), residuals 10^{-14} . Step T3: the level-3 exit rate from the branch is measured at two amplitudes and is linear in A ($39.32 \cdot A$ and $39.45 \cdot A$ at the two points); the closure consistency breaks in the first stage — the branch is dynamically empty.

Table 6.1. Layers of the verdict march_dae_nonlinear.json (both points)

Layer	Statement	Key numbers
T1_flat_line	the line = exact equilibria; the 8-echo march frozen	$F \leq 6.3 \times 10^{-13}$; drift 0.0 / 8.6×10^{-12}
T2_static_S	$e2$ does not touch S ; $b2$ holds the radius statically	0.0577 const; 1.0255–1.0383; $ r \leq 1.4 \times 10^{-15}$
T3_flow_exit	the exit from $b2$ is linear in A ; breakdown in stage 1	$39.32 \cdot A / 39.45 \cdot A$; $ r = 3.9 \times 10^{-5}$ – 3.9×10^{-4}
stats_machine	11 755 closure assemblies; 0 rank drops	min rank 9; max solv res 3.9×10^{-4}

5. Numerical Results

Table 6.2. The march along the flat line (eight echoes, RK4 + projections)

Point	Distance per echo	Max. drift	F_max	r_max
$\kappa = 2$	13.480672307	0.0	6.3×10^{-13}	0.0
$\kappa = 2 - \pi^2/98$	13.833463780	8.6×10^{-12}	3.1×10^{-11}	2.2×10^{-13}

Table 6.3. The exit from the b2 branch: linearity in the amplitude (both points)

A	$\ V\ $	exit $ DrV $	exit / A	march breakdown
0.001	0.0329 / 0.0335	0.03932 / 0.03935	39.32 / 39.35	$ r = 3.93 \times 10^{-5}$
0.010	0.3290 / 0.3353	0.39213 / 0.39453	39.21 / 39.45	$ r = 3.92 \times 10^{-4}$

The ratio exit/A is stable at 39.2–39.5 at both points and both amplitudes — the linearity in A is confirmed independently; the breakdown of the first march stage shows that the exit is not a weak asymptotics but an instantaneous inconsistency of the closure. The radius of the b2-branch is held statically up to $|x - x^*| \approx 1.35$ ($A = 0.05$: $1.359 \rightarrow 1.412$), that is, the emptiness of the branch is a dynamic fact, not a static non-existence.

6. Corollaries

Corollary 6.1 (the static core is exhausted). Jointly with the factorization (monograph No. 3) and the Krawczyk census (monograph No. 9), Theorem 6 closes the static picture: around x^* there is exactly one non-degenerate branch (the line), and it consists of equilibria. Any “live” near-critical dynamics must live outside the truncated DAE — in the PDE machine of v6–v9 and campaigns v20–v22.

Corollary 6.2 (the linear world is finally annihilated). The Jordan-2 block of v14 is not the linearization of any nonlinear flow; the vector b2, killed by the linear level-3, is not rescued in the nonlinear statics either. The “baryon asymmetry” programme on the tower is

closed: the $\pi/30$ residual can only be produced by finite-amplitude dynamics of the PDE level.

Corollary 6.3 (interpretation discipline). The exit rate (6.1) is linear in the amplitude and large ($\approx 39A$); any observed slow variables near x^* cannot be the coordinates of the second flows of the tower. This is the ground of the v16–v17 redirection: the figures carry states, not motion.

7. Honest Caveats

The theorem concerns the truncated tower with the derived dd-law in the T1c source form; the stiffness is a property of the truncation. Puiseux branches at the soft pole $45/8$ are not excluded (monograph No. 8): at crossings of representations branch jumps are possible, though they do not affect the claims T1–T3 themselves, which are verified at each point directly. Global components of S far from x^* were not sought (shooting/nsolve — the following campaigns): the theorem is local in the neighbourhood of the critical point. Finally, the hidden conditions are computed in the Procrustes-aligned basis of the left kernel; the formulation itself is basis-invariant, but the absolute $\sigma_{\min}$ numbers in the run logs are not (the erratum v18a [B3] must be accounted for when comparing absolute values).

8. Reproducibility

Table 6.4. Reproduction protocol for Theorem 6

Step	Command	Result
1	python3 march_dae_nonlinear.py	results/ march_dae_nonlinear.json (theorems T1–T3, both points)
2	python3 march_delta_mono.py	results/march_delta_mono.json (the Jordan-2 secular growth)
3	python3 hexcycle_dae.py	results/hexcycle_dae.json (the global predictor, v16)
4	python3 -m pytest tests/ -q	58 passed (the march tests)

Step	Command	Result
		included)

The running time is 53 s (runtime_s) on an ordinary laptop. The data correspond to commit 5b256cd; autonomous copies of the machines live in `einstein_direct/open9/machines/` with a private copy of the results.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/dae_nonlinear_core.py`; `einstein_direct/march_dae_nonlinear.py`; `einstein_direct/results/march_dae_nonlinear.json` (campaign v15, LEDGER §26).
- [2] Ibid.: `einstein_direct/sympy_dd_closure.py`; `einstein_direct/results/dd_closure.json` (the linear stub, v14).
- [3] Ibid.: `einstein_direct/hexcycle_dae.py`; `einstein_direct/results/hexcycle_dae.json` (the global predictor, v16).
- [4] Ibid.: `einstein_direct/README.md` §§1.3, 3.5; `einstein_direct/README_EN.md` §26.
- [5] Monographs No. 1, No. 3, No. 5, No. 8, No. 9 of the theorems12 cycle.